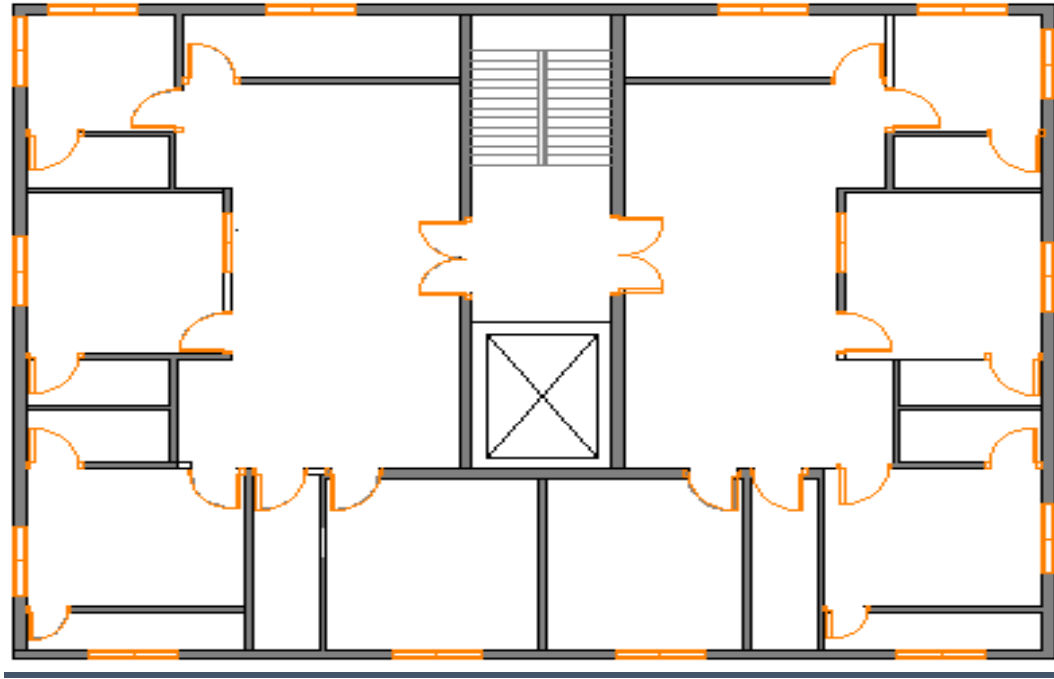


Electrical Design and Planning of a Multi-Storey Building



Course Title: **Electrical Wire & Drafting**

Submitted By:

Group Members (Total: 7)

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Objective of the Project

The primary objective of this project is to design and develop a complete civil and electrical system for a 4000 sq.ft, 6-storey residential building, ensuring safety, functionality, efficiency, and compliance with relevant standards and codes.

The specific objectives are:

- To prepare the civil architectural and structural layout of the building, including floor plans, load considerations, and space utilization.
- To design fitting and fixture layouts, ensuring proper placement of lighting, switches, sockets, and appliances for user convenience and energy efficiency.
- To develop the electrical conduit layout, providing safe, systematic, and concealed wiring routes.
- To perform SB (Switch Board) calculations based on connected load and to group SBs logically with proper connection diagrams.
- To calculate and design the SDB (Sub Distribution Board), including load distribution, protection devices, and connection diagrams.
- To perform MDB (Main Distribution Board) calculations for the entire building and prepare its detailed connection diagram
- To prepare a Single Line Diagram (SLD) representing the complete electrical power distribution system of the building.
- To design an effective earthing system, ensuring protection against electric shock and equipment damage.
- To plan and design the lightning protection system, safeguarding the building from lightning-induced hazards.
- To develop the fire detection and safety system layout, ensuring early detection and compliance with safety regulations.

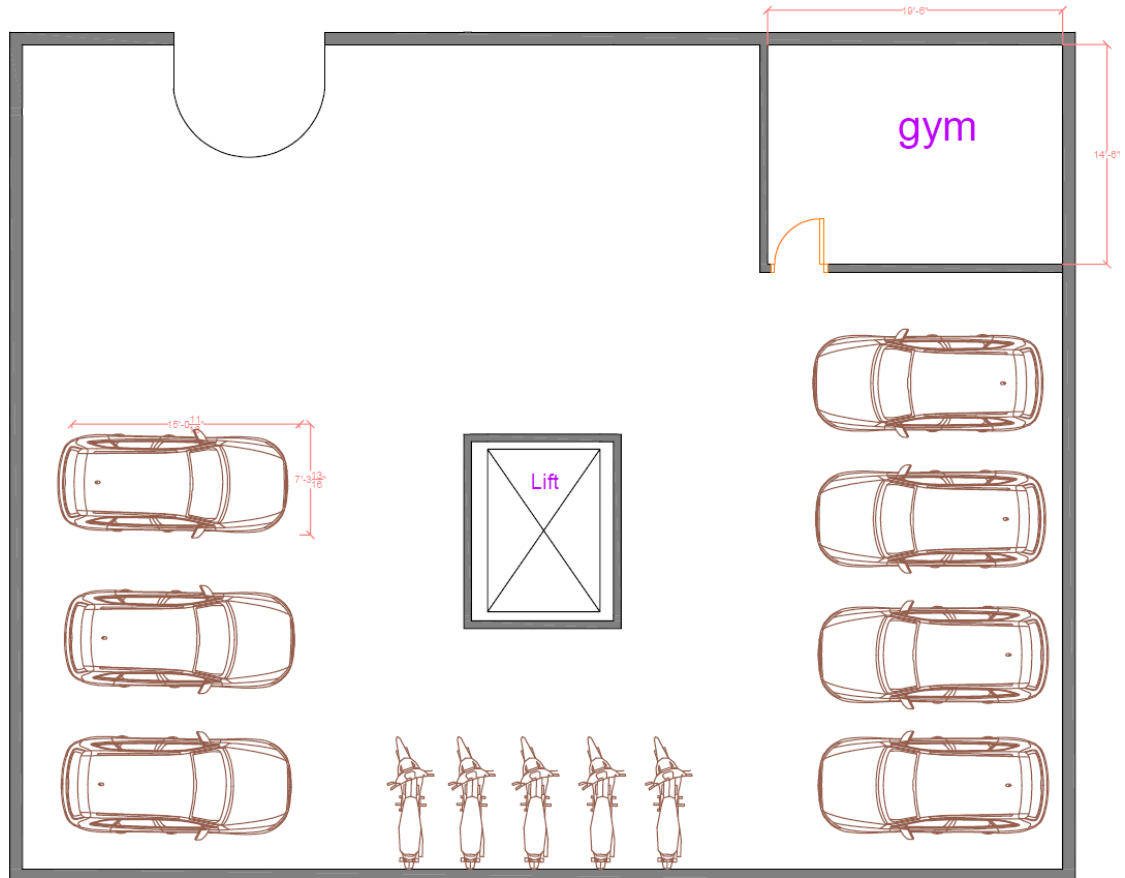
Design Steps Of The project

The project was carried out according to the following design steps:

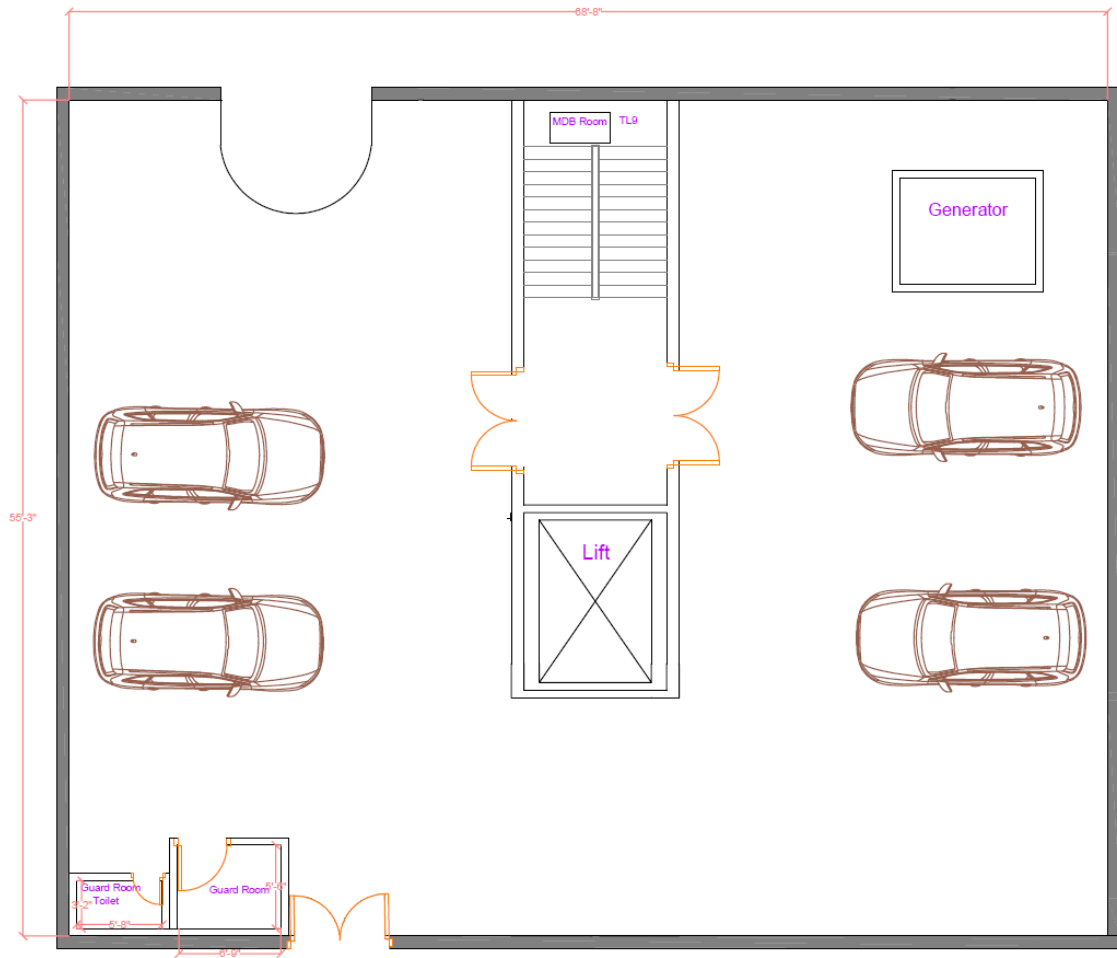
- ❖ Ground floor and typical floor plan of a three-storey building.
- ❖ Fittings and fixtures for each floor.
- ❖ Conduit layout planning for each floor.

- ❖ Switchboard and distribution board diagram.
- ❖ Lightning protection system (LPS) design.

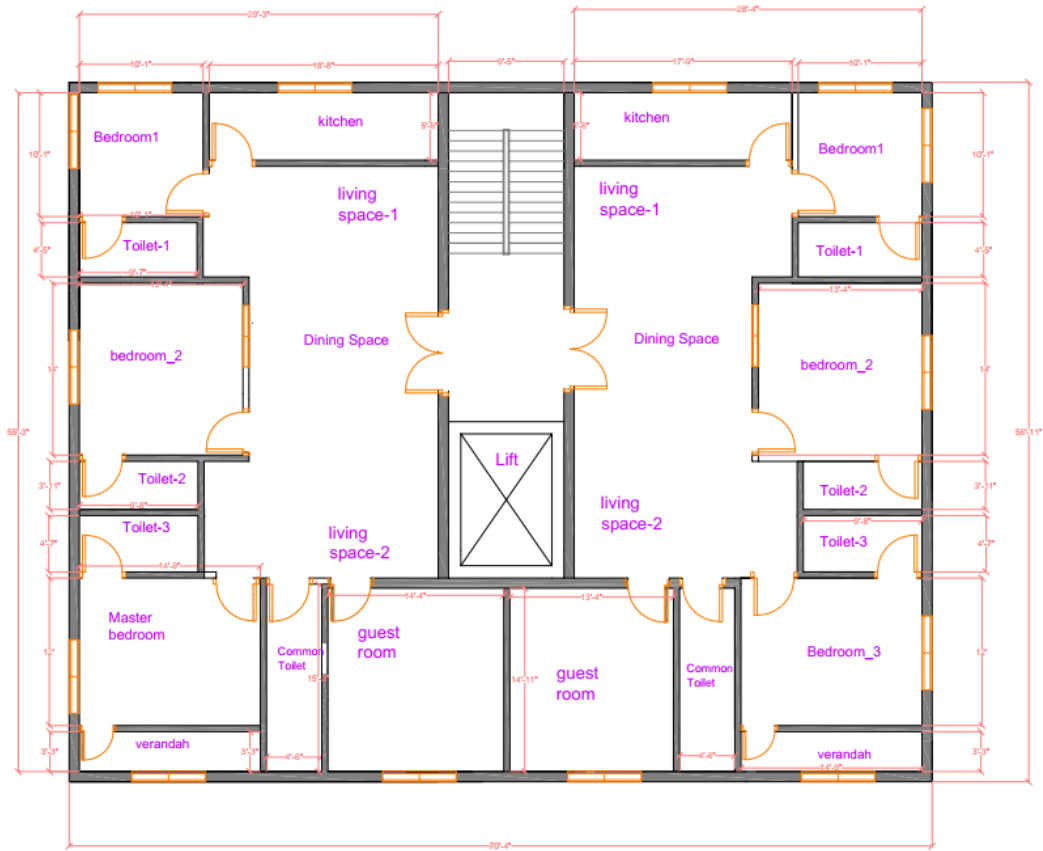
Basement Floor:



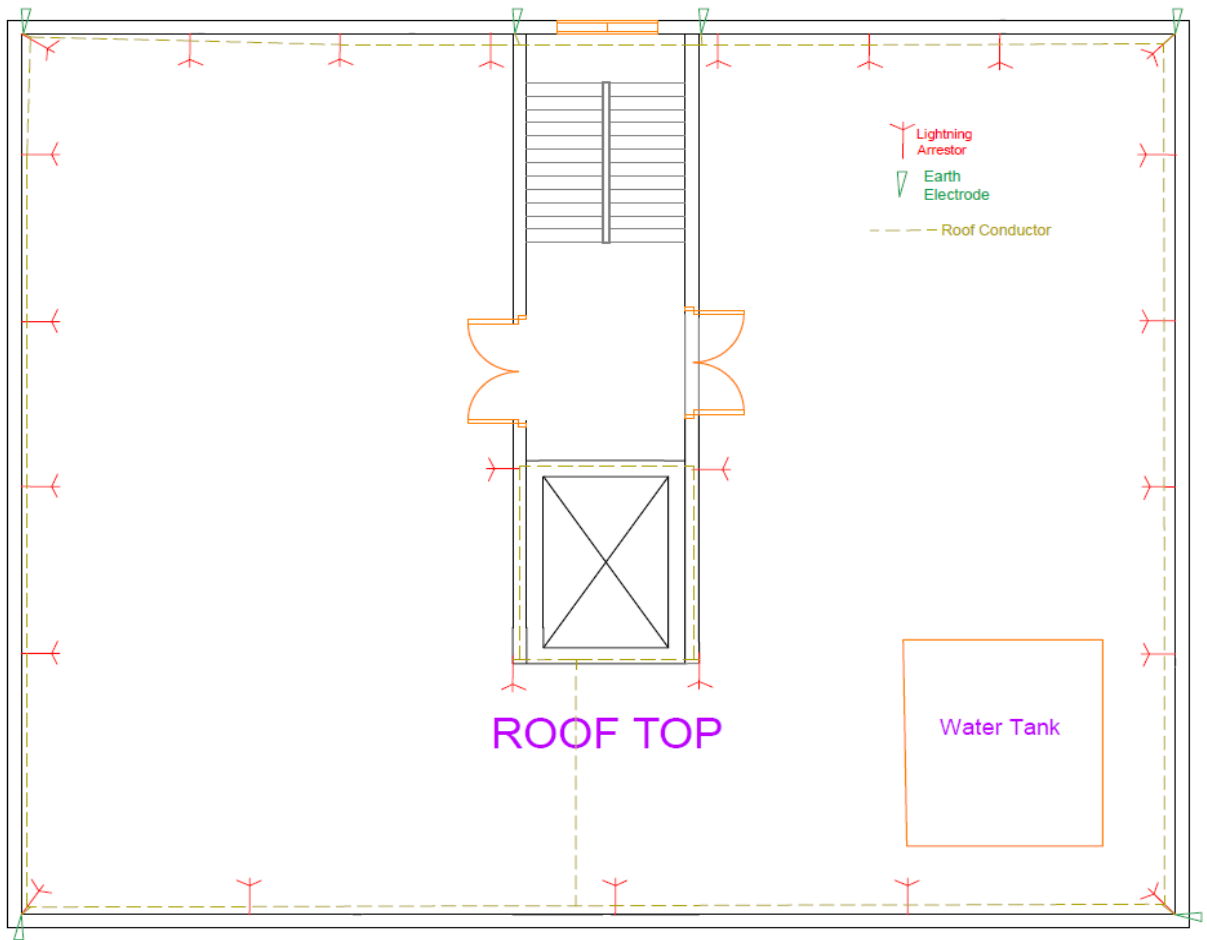
Ground Floor:



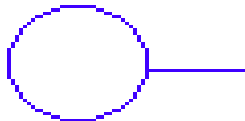
Typical Floor:

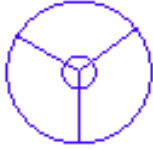
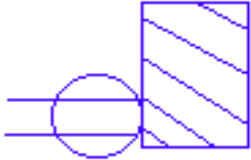
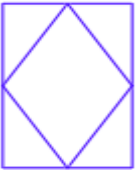
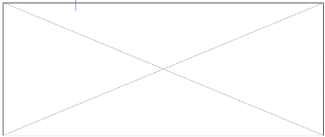
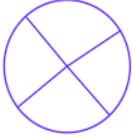


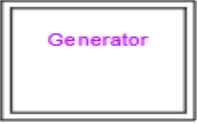
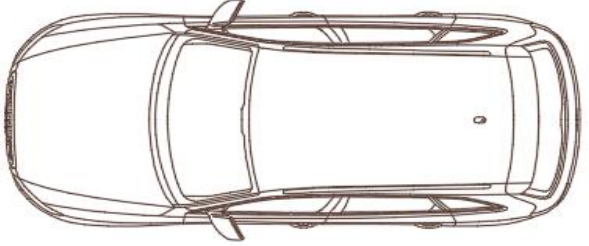
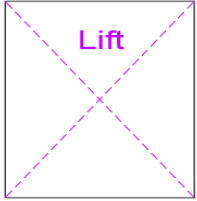
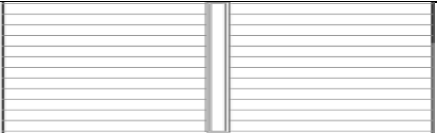
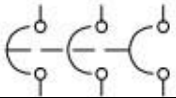
Rooftop:



FIXTURE LEGENDS

Description	Caption	Symbol
LED light	LL	
Tube Light	TL	

Fan	F	
Switch Board	SB	
Sub Distribution Board	SDB	
Main Distribution Board	MDB	
Exhaust Fan	E	
3 Pin Socket 20A	Socket	
Door large	D_large	
Door Small	D_small	
Window large	W_large	
Window small	W_small	

Generator	G	
Car Symbol	Audi	
Bike symbol	Bike	
Lift Symbol	lift	
Stairs	s	
Water Tank	T	
Door Bell	Bell	
Triple Pole Circuit Breaker (TP MCCB)	MCCB,MCB	

Calculation part

Theory :

For Light

Let,

Room length = L (in meters)

width = W (in meters)

N = Number of lights required

E = Luminance level required (lux). This parameter will vary depending on the type of room (e.g. bedroom, kitchen)

F = Average luminous flux from each light source (lumen)

UF = Utilization factor (allowance for light distribution of the luminaire and the room surfaces)

MF = Maintenance factor (allowance for reduced light output due to deterioration)

Then, following is the equation used to calculate the number of lights required ^[1]:

$$N = \frac{E * L * W}{F * UF * MF}$$

The following table shows the required luminance values for each room.

Room Type	E (lux)
Dining Space	110
Living Room	110
Kitchen	215
Bedroom-1	215
Bedroom-2	215
Guest Bedroom	215
Veranda	215
Gym Room	215
Bathroom	215
Basement	100
Guard Room	215

Calculation of Utilization Factor:

Table 1: Utilisation Factors

Room reflectances			Room index								
C	W	F	0.75	1.00	1.25	1.50	2.00	2.50	3.00	4.00	5.00
0.7	0.5	0.2	NA	0.61	0.65	0.67	0.70	0.71	0.73	0.74	0.75

$$\text{Room Index} = \frac{L * W}{\text{Mounting Height} * (L + W)}$$

Fan Requirement:

The number of fans required, M is determined by the following formula:

$$M = \frac{L(\text{in ft}) * W(\text{in ft})}{100}$$

Environment	Lux
Living Room	108–215
Bedrooms	215–538
Kitchen-General	215–538
Kitchen-Preparing/Cooking	538–1,076
Dining Room	108–215
Rooms for Difficult Reading & Writing	538–1,076
Bathrooms	215–538
Hallways	54–108
Basement/Garage Workshop	215–807
Basement-Laundry	215–1,076
Basement-Exercise and Recreation	323

LED TUBES OUTPUT		
Fluorescent Equivalent	LED Wattage	Lumens
28W	12W	1,715 lm
32W	14W	1,920 lm
35W	15W	2,172 lm
40W	18W	2,567 lm



Room	E (lux)	Height (ft)	Width (ft)	Area (m ²)	UF	N (Light)	M (Fan)
Bedroom 1	215	10.083	10.083	9.45	0.75	1.05 (1)	1.01 (1)
Toilet 1	215	4.42	9.58	3.9	0.75	0.77 (1)	
Bedroom 2	215	14	13.33	17.34	0.75	1.94 (2)	1.87 (2)
Toilet 2	215	3.92	9.67	3.52	0.75	0.695 (1)	
Master Bedroom	215	12	14.75	16.44	0.75	1.84 (2)	1.77 (2)
Toilet 3	215	4.58	9.67	4.11	0.75	0.81 (1)	
Guest room	215	14.92	14.33	19.86	0.75	2.22 (3)	2.13 (2)
Common toilet	215	15.25	4.5	6.37	0.75	0.91 (1)	
Living space 1	110	19	8.83	54.4	0.75	3.1 (3)	5.87 (4)
Dining space		15.83	15				
Living space 2		19.08	9.5				
Veranda	215	3.25	3.25	0.98	0.75	0.19 (1)	
Kitchen	215	5.5	18.67	9.54	0.75	1.88 (2)	1.02 (1)

Bedroom-1:

$$L = 10 + 1/12 = 10.083 \text{ ft}$$

$$W = 10 + 1/12 = 10.083 \text{ ft}$$

$$A = 10.083 \times 10.083 = 101.67 \text{ Sq ft}^2 = 101.67 \times (0.3048)^2 = 9.45 \text{ Sq m}^2$$

$$\text{Lumens, } F = 2567 \text{ lm; Lux, } E = 215; \text{ LLF} \times \text{UF} = 0.75$$

$$\text{Number of Light} = E \times A / F \times \text{LLF} \times \text{UF} = (215 \times 9.45) / (2567 \times 0.75) = 1; (\text{LED Tubes } 40 \text{ W})$$

$$\text{Number of Fan} = A / 100 = 101.67 / 100 = 1$$

Toilet-1:

$$L = 4 + 5/12 = 4.42 \text{ ft}$$

$$W = 9 + 7/12 = 9.58 \text{ ft}$$

$$A = 4.42 * 9.58 = 42.34 \text{ Sq ft}^2 = 3.9 \text{ Sq m}^2$$

$$\text{Lumens, } F = 1450 \text{ lm; Lux, } E = 215; \text{LLF} * \text{UF} = 0.75$$

$$\text{Number of Light} = (215 * 3.9) / (1450 * 0.75) = 0.77 \sim 1$$

(15W LED Bulb)

Bedroom-2:

$$L = 14 \text{ ft; } W = 13 + 4/12 = 13.33 \text{ ft}$$

$$A = 14 * 13.33 = 186.62 \text{ Sq ft}^2 = 17.34 \text{ Sq m}^2$$

$$\text{Number of Light} = E * A / F * \text{LLF} * \text{UF} = (215 * 17.34) / (2567 * 0.75) = 1.94 \sim 2$$

(40W LED Tubes)

$$\text{Number of Fan} = 186.62 / 100 = 1.87 \sim 2$$

Toilet-2:

$$L = 3 + 11/12 = 3.92 \text{ ft}$$

$$W = 9 + 8/12 = 9.67 \text{ ft}$$

$$A = 3.92 * 9.67 = 37.91 \text{ Sq ft}^2 = 3.52 \text{ Sq m}^2$$

$$\text{Number of Light} = E * A / F * \text{LLF} * \text{UF} = (215 * 3.52) / (1450 * 0.75) = 0.695 \sim 1$$

Master Bedroom:

$$L = 12 \text{ ft}$$

$$W = 14 + 9/12 = 14.75 \text{ ft}$$

$$A = 12 * 14.75 = 177 \text{ Sq ft}^2 = 16.44 \text{ Sq m}^2$$

$$\text{Lumens, } F = 2567; \text{ Lux, } E = 215;$$

$$\text{Number of Light} = (215 * 16.44) / (2567 * 0.75) = 1.84 \sim 2 \text{ (40W LED Tubes)}$$

$$\text{Number of Fan} = 177/100 = 1.77 \sim 2$$

Toilet-3:

$$L = 4 + 7/12 = 4.58 \text{ ft}$$

$$W = 9.67 \text{ ft}$$

$$A = 4.58 * 9.67 = 44.29 \text{ Sq ft}^2 = 4.11 \text{ Sq m}^2$$

$$\text{Light} = (215 * 4.11) / (1450 * 0.75) = 0.81 \sim 1; \text{ (15W LED Bulb)}$$

Common Toilet:

$$L = 15 + 3/12 = 15.25 \text{ ft}$$

$$W = 4.5 \text{ ft}$$

$$A = 68.63 \text{ Sq ft}^2 = 6.37 \text{ Sq m}^2$$

$$\text{Light} = (215 * 6.37) / (2000 * 0.75) = 0.91 \sim 1; \text{ (LED Bulb, (15-20) W} \rightarrow \text{Lumens, } F = 2000)$$

Guest Room:

$$L = 14 + 11/12 = 14.92 \text{ ft}$$

$$W = 14 + 4/12 = 14.33 \text{ ft}$$

$$A = 213.8 \text{ Sq ft}^2 = 19.86 \text{ Sq m}^2$$

$$\text{Light} = (215 * 19.86) / (2567 * 0.75) = 2.22 \sim 3 \text{ Light needed; (40W LED Tubes)}$$

$$\text{Fan} = 213.8 / 100 = 2$$

Verandha:

$$L = 3 + 3/12 = 3.25 \text{ ft}$$

$$W = 3.25 \text{ ft}$$

$$A = (3.25)^2 = 10.56 \text{ Sq ft}^2 = 0.98 \text{ Sq m}^2$$

$$\text{Light} = (215 * 0.98) / (1450 * 0.75) = 0.19 \sim 1 \text{ Light needed; (15W LED Bulb)}$$

Dining & Living Space:

$$L1 = 19 \text{ ft; } W1 = 8 + 10/12 = 8.83 \text{ ft (Living Space)}$$

$$L2 = 15 + 4/12 = 15.83 \text{ ft; } W2 = 15 \text{ ft (Dining Space)}$$

$$L3 = 19 + 1/12 = 19.08 \text{ ft; } W3 = 9.5 \text{ ft (Living Space)}$$

$$A = (19 * 8.83) + (15.83 * 15) + (19.08 * 9.5) = 586.5 \text{ Sq ft}^2 = 54.4 \text{ Sq m}^2$$

$$\text{Light} = (110 * 54.4) / (2567 * 0.75) = 3.1 \sim 3; (40W LED Tubes)$$

$$\text{Fan} = 586.5 / 100 = 5.87$$

Note: Dining space Fan needed~2 & we will give 1 Fan on the other 2 Living space.

Kitchen:

$$L = 5.5 \text{ ft}$$

$$W = 18 + 8/12 = 18.67 \text{ ft}$$

$$A = 5.5 \times 18.67 = 102.69 \text{ Sq ft}^2 = 9.54 \text{ Sq m}^2$$

$$\text{Light} = E \cdot A / F \cdot \text{LLF} \cdot \text{UF} = (215 \times 9.5) / (1450 \times 0.75) = 1.88 \sim 2$$

$$\text{Fan} = 102.69 / 100 = 1 \text{ (15W LED Bulb)}$$

Gym:

$$L = 19.5 \text{ ft}$$

$$W = 14.5 \text{ ft}$$

$$A = 19.5 \times 14.5 = 282.75 \text{ Sq ft}^2 = 26.3 \text{ Sq m}^2$$

$$\text{Lumens, } F = 2567; \text{ Lux} = 215; \text{ LED Tubes} = 40\text{W}$$

$$\text{Light} = (215 \times 26.3) / (2567 \times 0.75) = 2.9 \sim 3$$

$$\text{Fan} = A / 100 = 282.75 / 100 = 3$$

Basement:

$$L = 68 + 6.5 / 12 = 68.54 \text{ ft}$$

$$W = 54 + 1.56 / 12 = 54.13 \text{ ft}$$

$$A = 68.54 \times 54.13 = 3710.07 \text{ Sq ft}^2 = 3710.07 \times (0.3048) = 344.67 \text{ Sq m}^2$$

$$\text{Number of lights} = (100 \times 344.67) / (2567 \times 0.75) = 18; \text{ (Assume Lux} = 100, \text{ for standard basement)}$$

Ground:

$$L = 55 + 3 / 12 = 55.25 \text{ ft}$$

$$W = 68 + 8 / 12 = 68.67 \text{ ft}$$

$$A = 55.25 \times 68.67 = 3797.451 \text{ Sq ft}^2 = 352.79 \text{ Sq m}^2$$

$$\text{Lumen} = 2567; \text{ Lux} = 50; \text{ LED Tubes} = 40\text{W}$$

$$\text{Light} = (50 \times 352.79) / (2567 \times 0.75) = 9.1 \sim 9$$

Guard Room:

$$L = 5.5 \text{ ft}$$

$$W = 6 + 9/12 = 6.75 \text{ ft}$$

$$A = 37.13 \text{ Sq ft}^2 = 3.45 \text{ Sq m}^2$$

$$\text{Lumen} = 1450; \text{Lux} = 215; 15\text{W LED Light}$$

$$\text{Light} = (215 \times 3.45) / (1450 / 0.75) = 0.68 \sim 1$$

$$\text{Fan} = 37.13 / 100 = 1 \text{ fan needed}$$

Guard Room Toilet:

$$L = 3 + 2/12 = 3.17 \text{ ft}$$

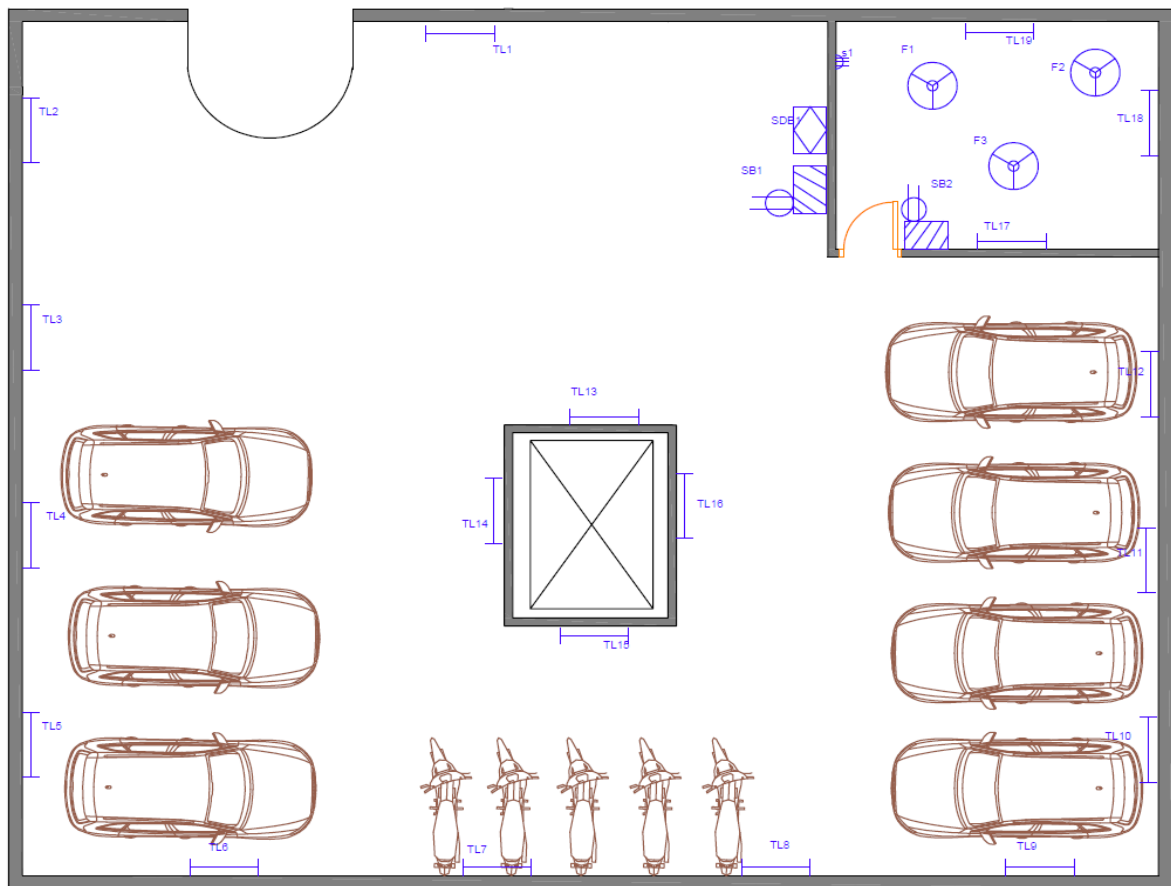
$$W = 5 + 8/12 = 5.67 \text{ ft}$$

$$\text{Lumen} = 1450; \text{Lux} = 215$$

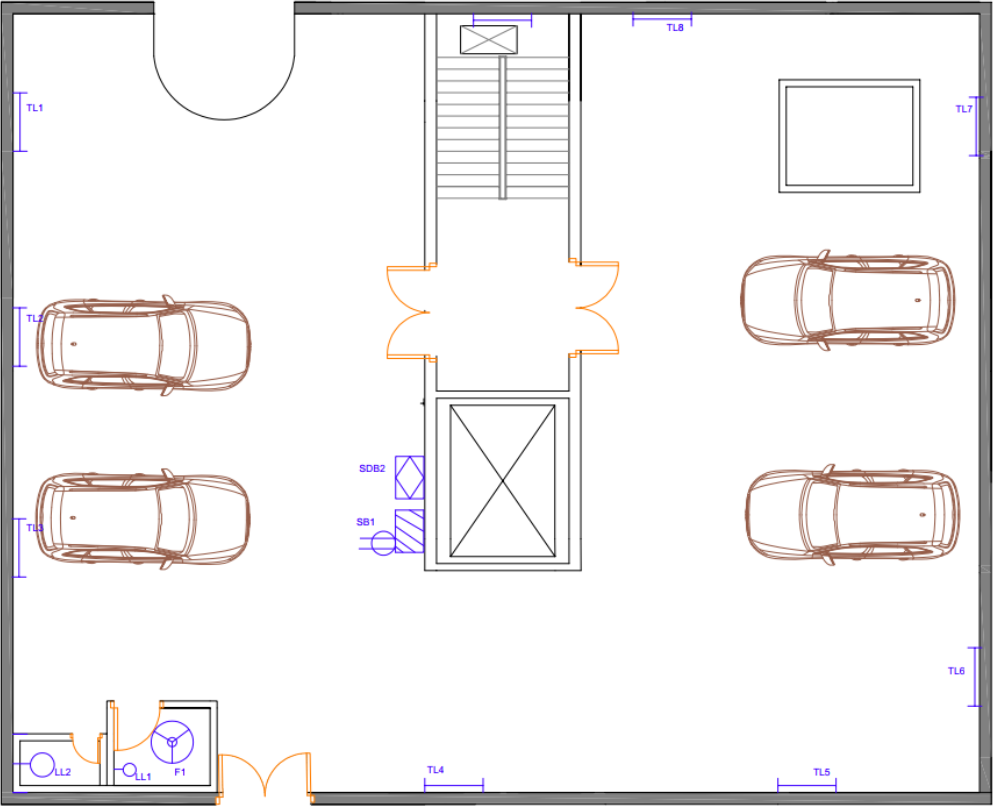
$$A = 17.97 \text{ Sq ft}^2 = 0.51 \text{ Sq m}^2$$

$$\text{Light} = (215 \times 0.51) / (1450 \times 0.75) = 0.1 \sim 1$$

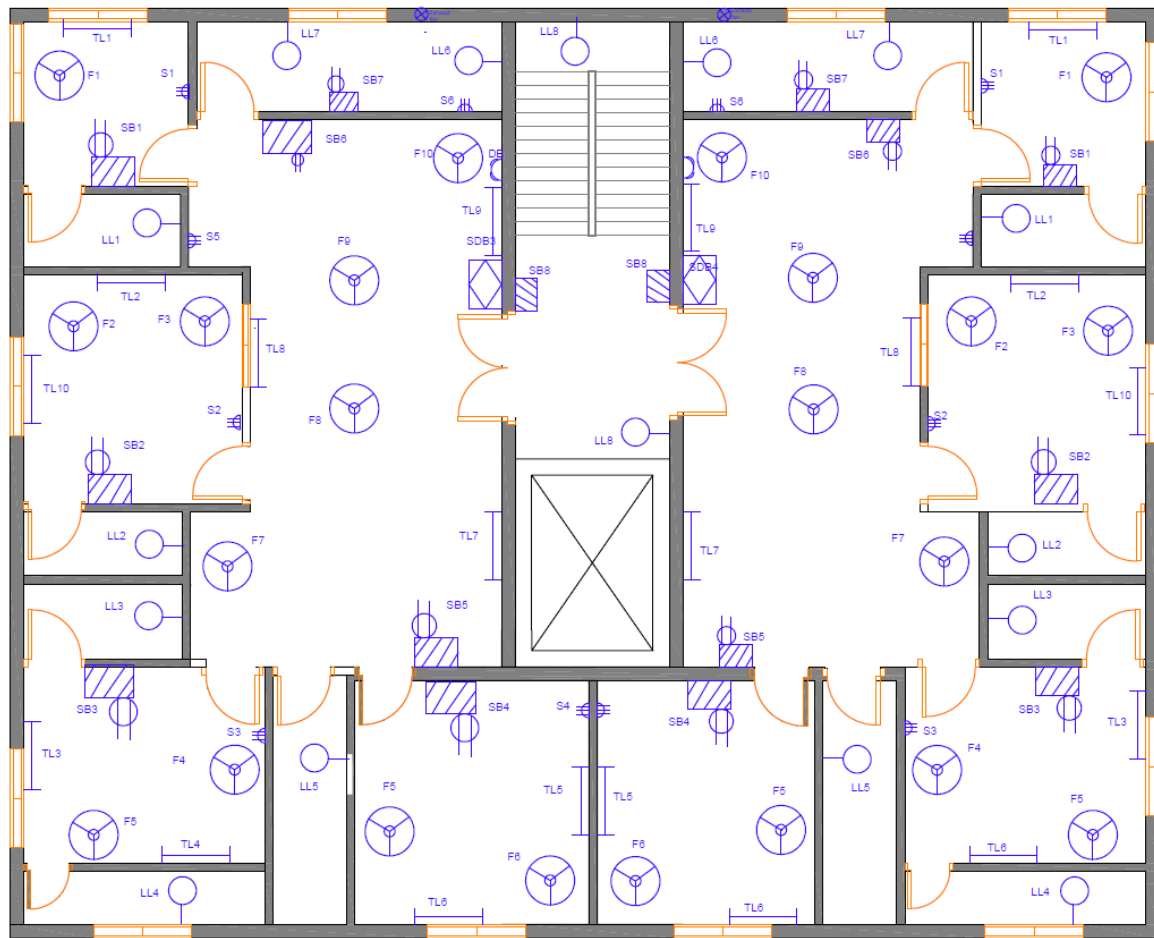
Basement Floor Fixtures:



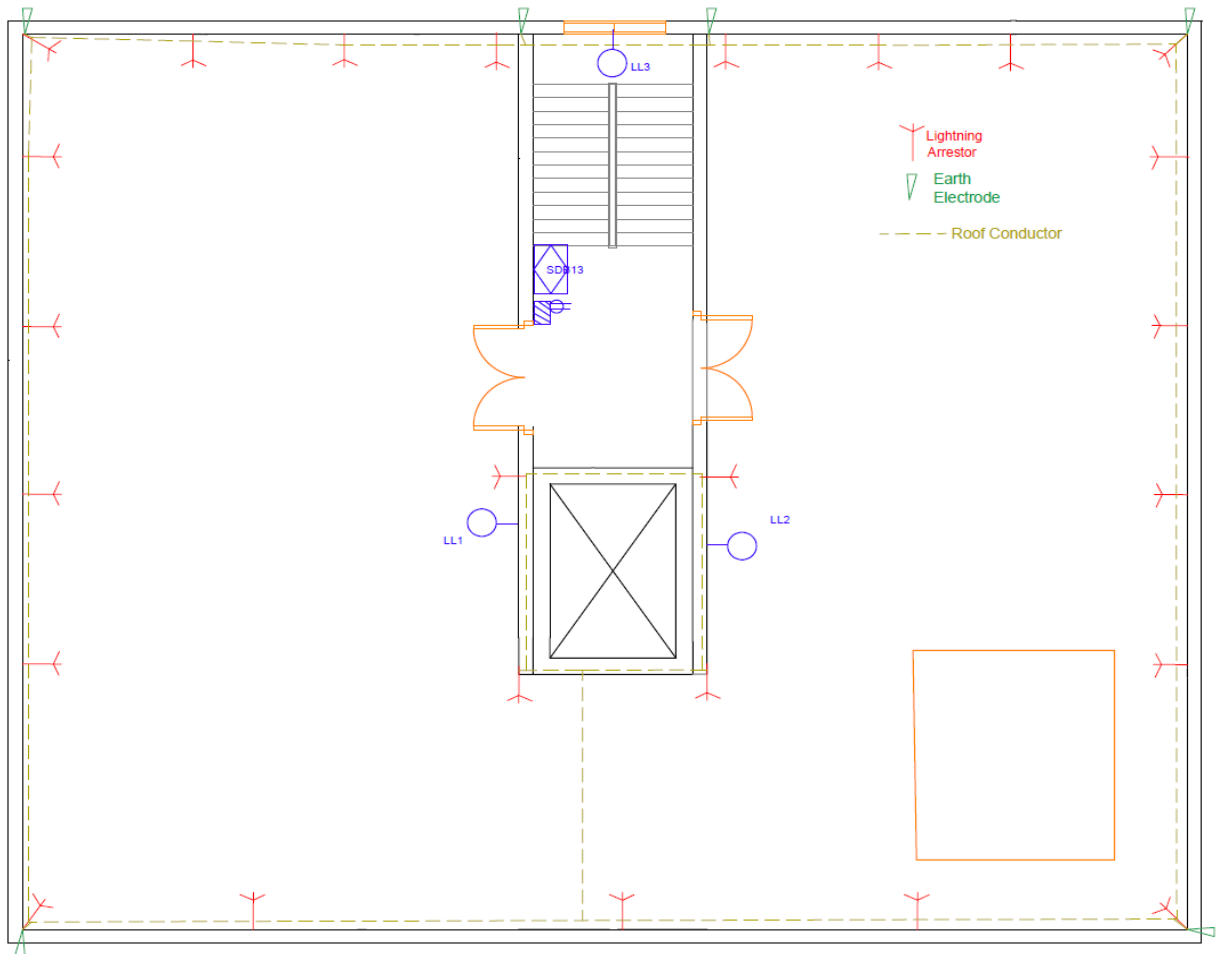
Ground Floor Fixtures:



Typical Floor :



Rooftop:

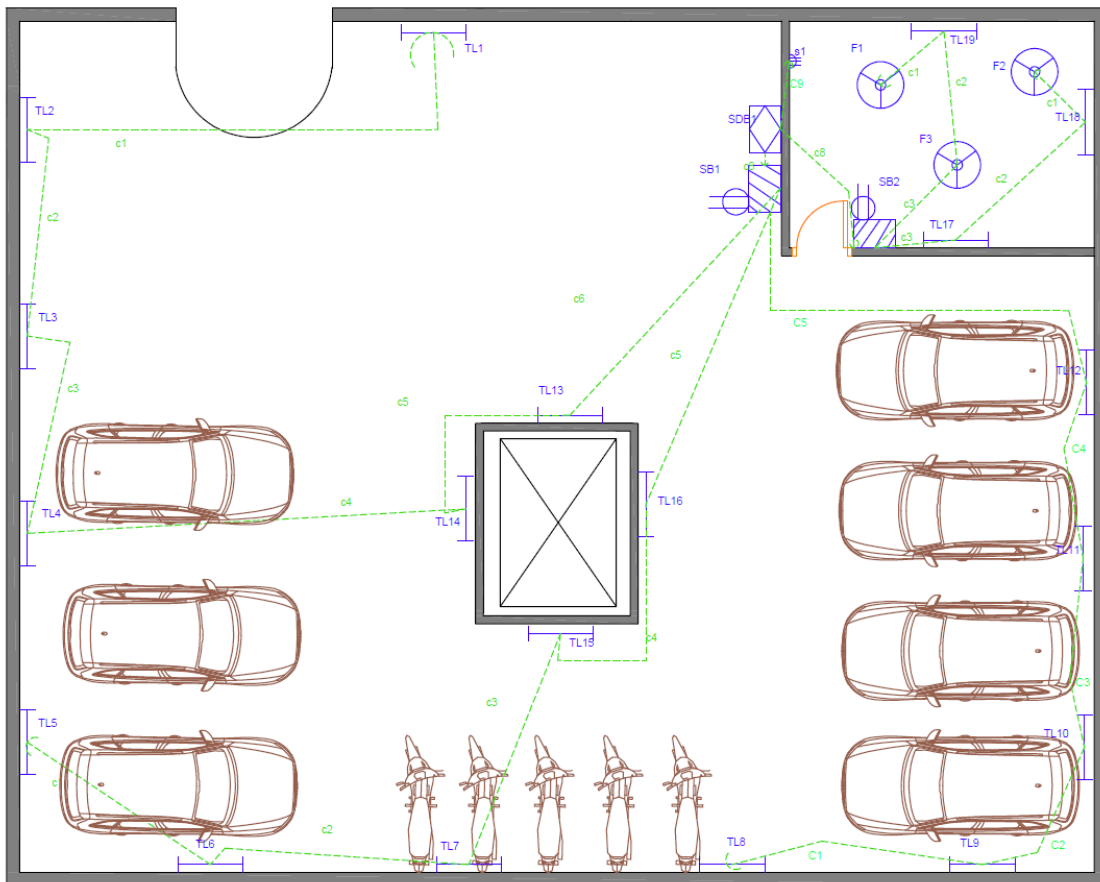


Conduit

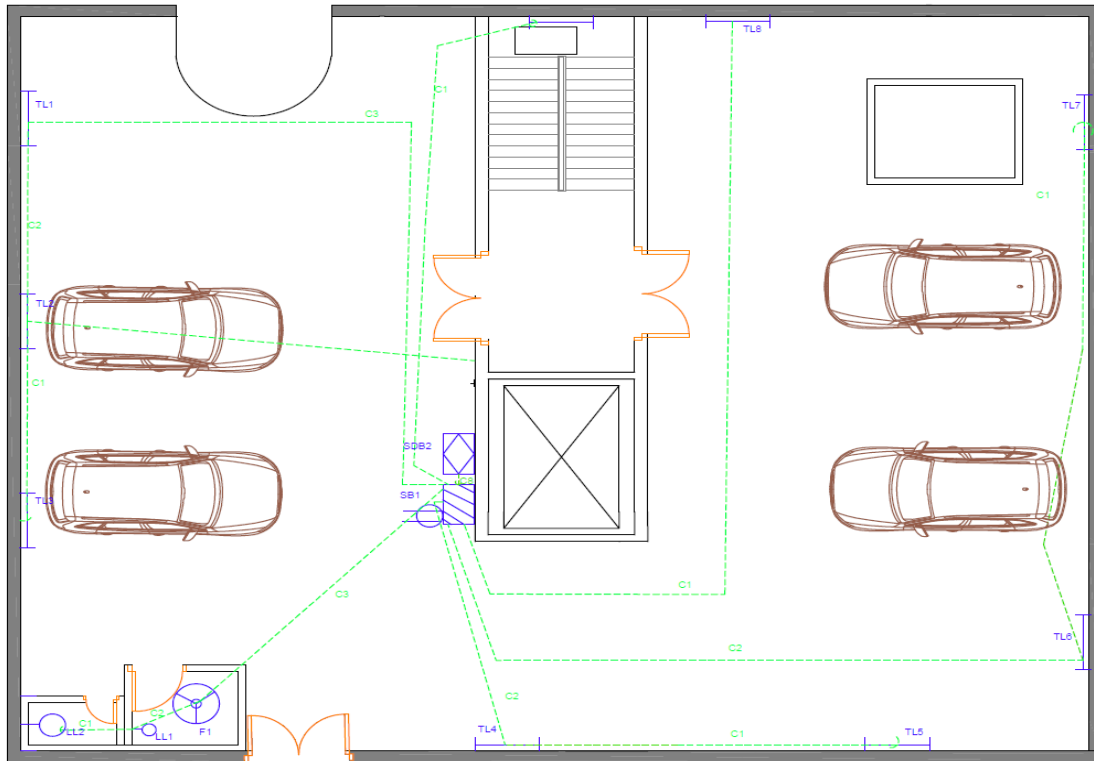
LEGENDS (WIRE RATING)

Name	Description	Current Rating	Conduit Diameter
C1	2 X 1.5rm BYM	5A	3/4"
C2	4 X 1.5rm BYM	5A	3/4"
C3	6 X 1.5rm BYM	5A	3/4"
C4	8 X 1.5rm BYM	5A	1"
C5	10 X 1.5rm BYM	5A	1"
C8	2 X 4rm BYM + 4rm BYA ECC	15A	3/4"
C9	2 X 6rm BYM + 6rm BYA ECC	20A	1"

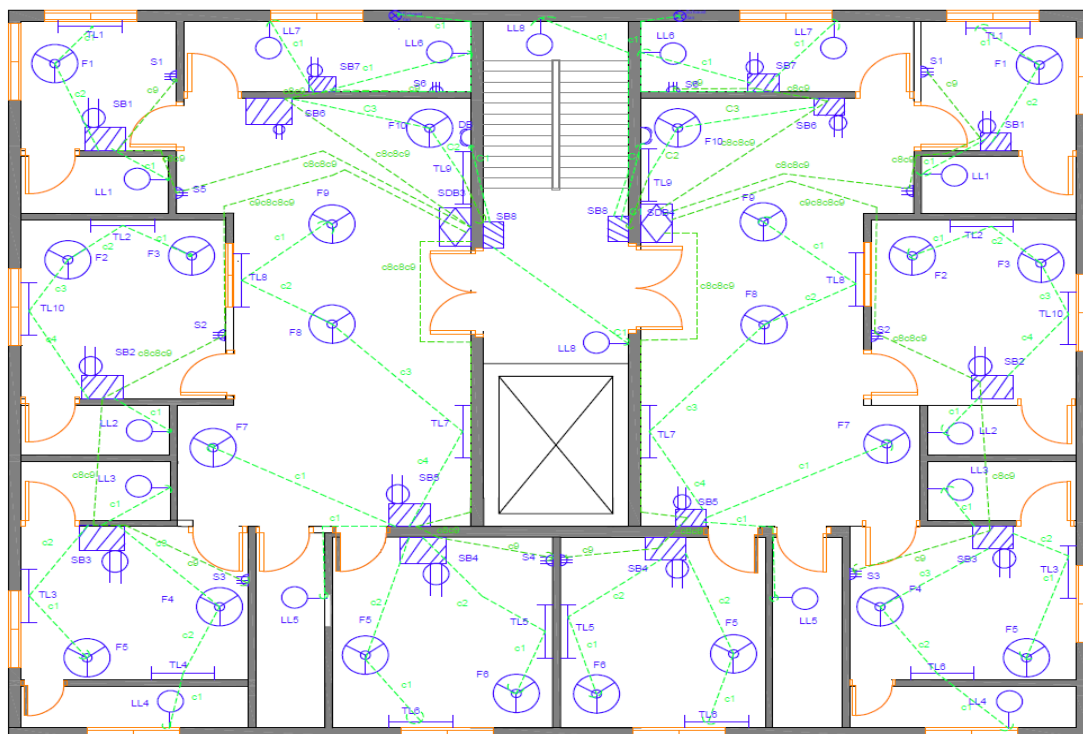
Basement :



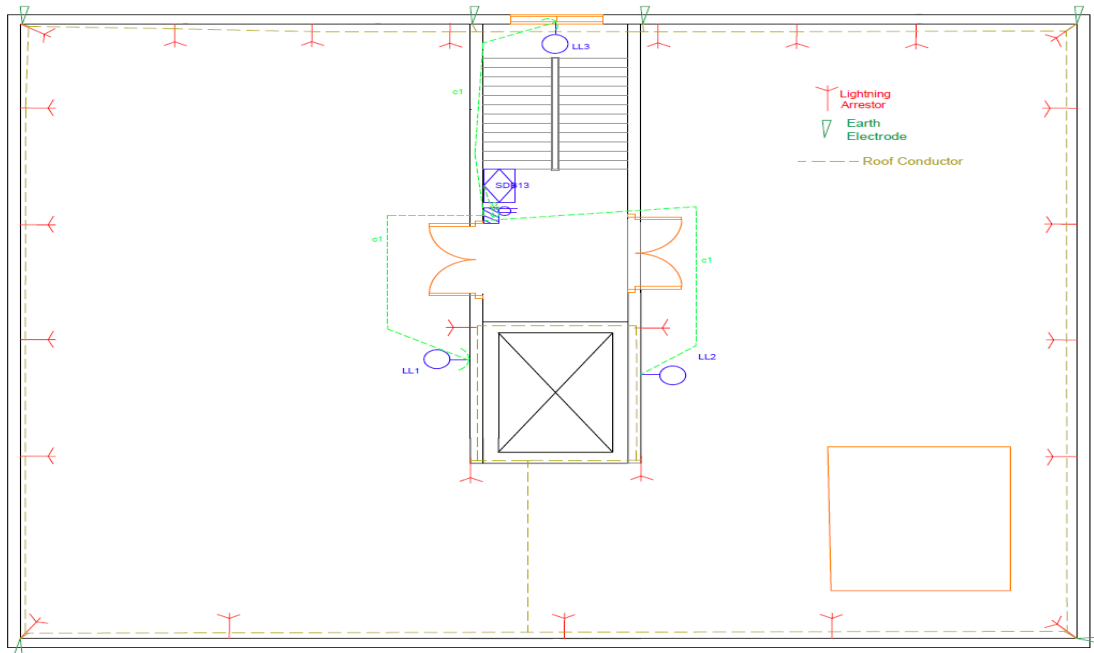
Ground floor:



Typical Floor:

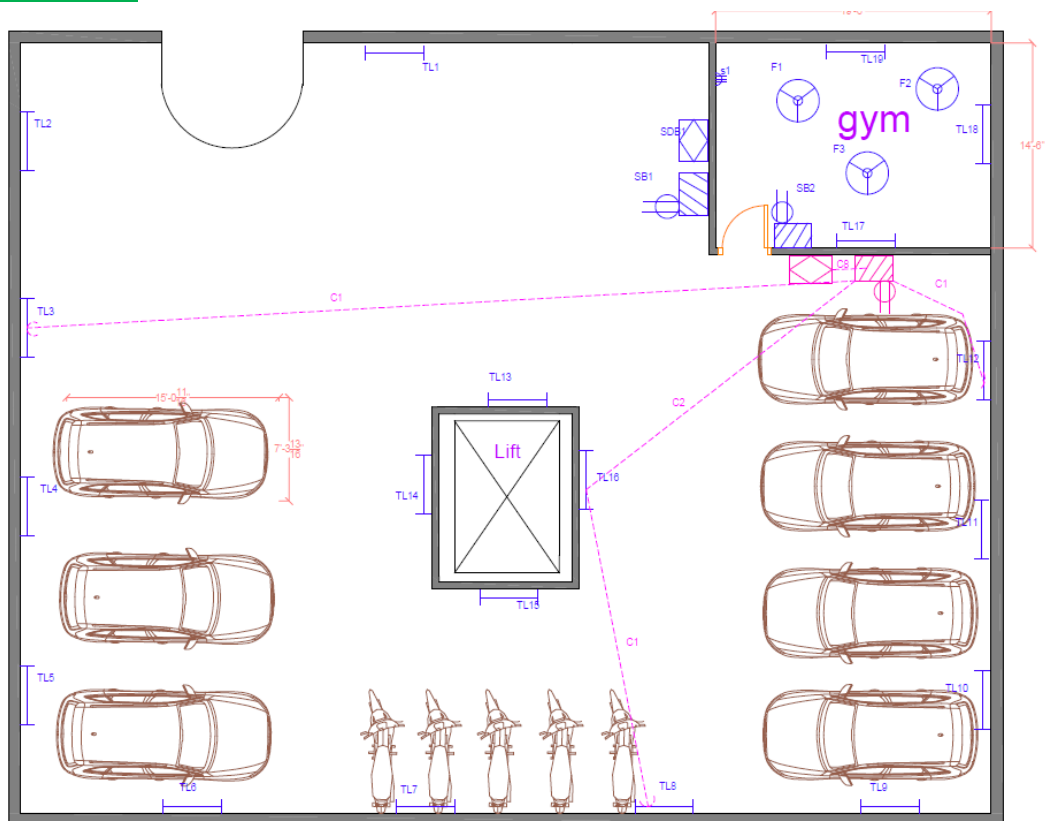


Roof top:

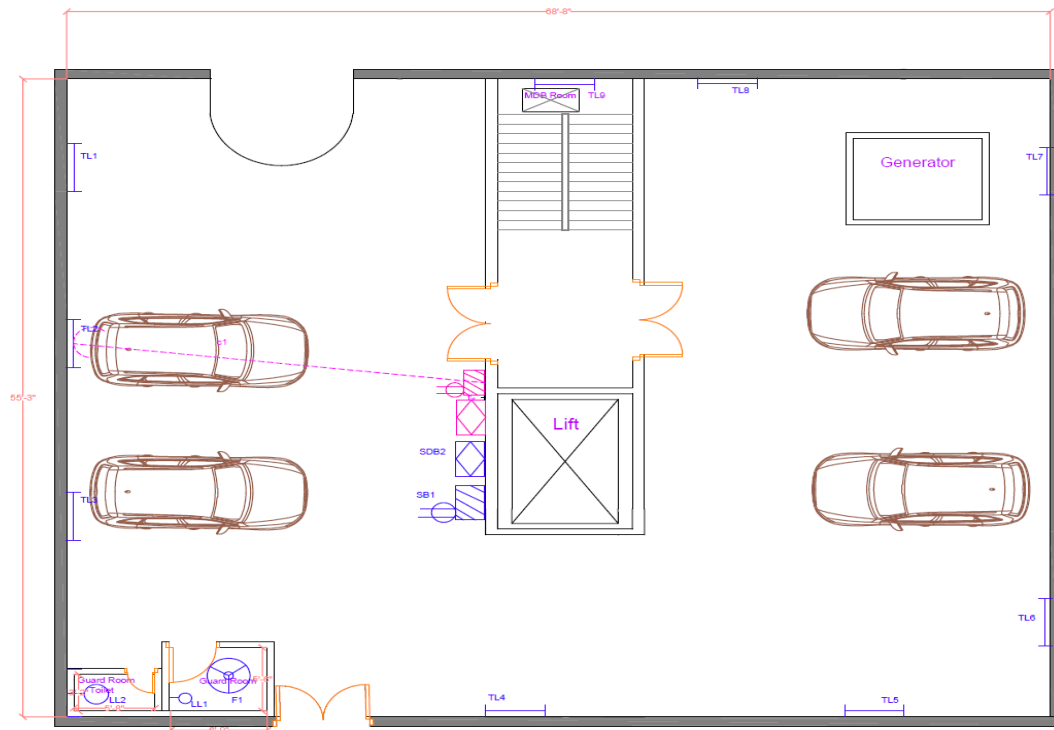


Emergency Conduit

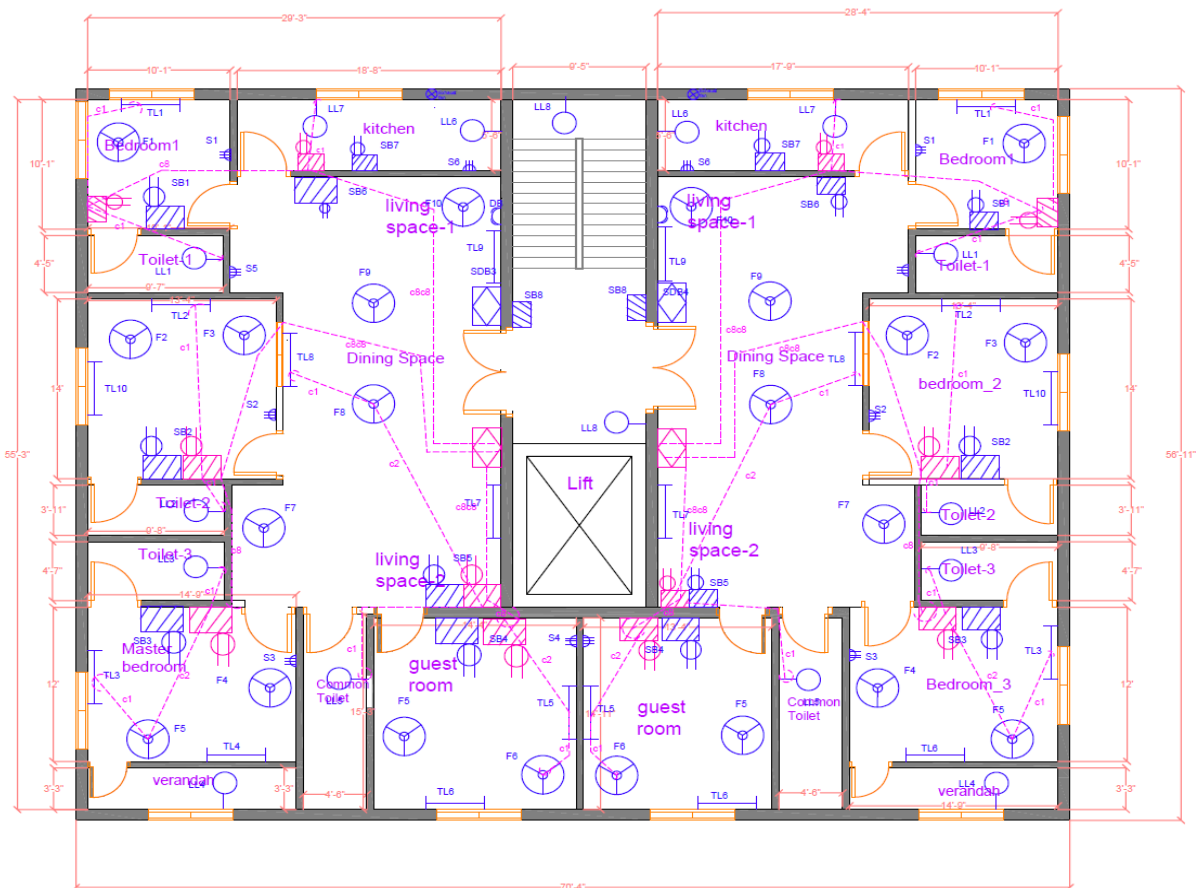
Basement :



Ground:



Typical floor:



Swtich Board

Calculation:

Switch Board Calculation (Level 1-)		
Switch Board – 1 (Bedroom 1+Toilet 1) 1L (40W) + 1L (15W) = 55 W 1F = 30 W 1X5A socket Total P = 85 W $P = V.I.\cos \theta$ Here, $V = 220V$ Let, $p.f = 0.95$ Now, $I = (P) / (V \cos \theta)$ = 0.40 Amp	Switch Board – 2 (Bedroom 2+Toilet2) 2L (40W) + 1L (15W) = 95 W 2F = 60 W 1X5A socket Total = 155 W $P = V.I.\cos \theta$ Here, $V = 220V$ Let, $p.f = 0.95$ Now, $I = (P) / (V \cos \theta)$ = 0.74 Amp	Switch Board – 3 (Master Bedroom+ Toilet 3 + Verandah) 2L (40W) + 2L (15W) = 110 W 2F = 60 W 1X5A socket Total = 170 W $P = V.I.\cos \theta$ Here, $V = 220V$ Let, $p.f = 0.95$ Now, $I = (P) / (V \cos \theta)$ = 0.81 Amp
Switch Board – 4 (Guest room) 2L (40W) = 80 W 2F = 60 W 1X5A socket Total = 140 W $P = V.I.\cos \theta$ Here, $V = 220V$ Let, $p.f = 0.95$ Now, $I = (P) / (V \cos \theta)$ = 0.66 Amp	Switch Board – 5 (Living space 2+ Common Toilet) 2L (40W) + 1L (15W)= 95 W 3F = 90 W 1X5A socket Total = 185 W $P = V.I.\cos \theta$ Here, $V = 220V$ Let, $p.f = 0.95$ Now, $I = (P) / (V \cos \theta)$ = 0.88 Amp	Switch Board – 6 (Living space 1) 1L (40W) = 40 W 1F = 30 W 1X5A socket Total = 70 W $P = V.I.\cos \theta$ Here, $V = 220V$ Let, $p.f = 0.95$ Now, $I = (P) / (V \cos \theta)$ = 0.33 Amp
Switch Board – 7 (Kitchen) 2L = 30 W 1F = 40 W 1X5A socket Total = 70 W $P = V.I.\cos \theta$ Here, $V = 220V$ Let, $p.f = 0.95$ Now, $I = (P) / (V \cos \theta)$ =0.33 Amp	Switch Board – 8 (Bedroom 1+Toilet 1) 1L (40W) + 1L (15W) = 55 W 1F = 30 W 1X5A socket Total P = 85 W $P = V.I.\cos \theta$ Here, $V = 220V$ Let, $p.f = 0.95$ Now, $I = (P) / (V \cos \theta)$ = 0.40 Amp	Switch Board – 9 (Bedroom 2 +Toilet 2) 2L (40W) + 1L (15W) = 95 W 2F = 60 W 1X5A socket Total = 155 W $P = V.I.\cos \theta$ Here, $V = 220V$ Let, $p.f = 0.95$ Now, $I = (P) / (V \cos \theta)$ = 0.74 Amp

Switch Board – 10 (Master Bedroom + Toilet 3 + Verandah) 2L (40W) + 2L (15W) = 110 W 2F = 60 W 1X5A socket Total = 170 W P = V.I.cos θ Here, V= 220V Let, p.f = 0.95 Now, I = (P)/ (V cosθ) = 0.81 Amp	Switch Board – 11 (Guest room) 2L (40W) = 80 W 2F = 60 W 1X5A socket Total = 140 W P = V.I.cos θ Here, V= 220V Let, p.f = 0.95 Now, I = (P)/ (V cosθ) = 0.66 Amp	Switch Board – 12 (Living space 2+ Common Toilet) 2L (40W) + 1L (15W)= 95 W 3F = 90 W 1X5A socket Total = 185 W P = V.I.cos θ Here, V= 220V Let, p.f = 0.95 Now, I = (P)/ (V cosθ) = 0.88 Amp
Switch Board – 13 (Living space 1) 1L (40W) = 40 W 1F = 30 W 1X5A socket Total = 70 W P = V.I.cos θ Here, V= 220V Let, p.f = 0.95 Now, I = (P)/ (V cosθ) = 0.33 Amp	Switch Board – 14 (Kitchen) 2L = 30 W 1F = 40 W 1X5A socket Total = 70 W P = V.I.cos θ Here, V= 220V Let, p.f = 0.95 Now, I = (P)/ (V cosθ) =0.33 Amp	

Switch Board Summary:

Sub distribution board (SDB)-1 Fixtures							
Room Name	Circuit No	Switchboard	Fixture	Power (W)	Current Rating(A)	Wire Rating	Breaker to SDB
Bedroom 1+Toilet 1	CKT1	SB1	TL1	40	0.44	C1&C2	10A
			LL1	15			
			F1	30			
			S1		5	C9	
			Total		5.40		
Bedroom 2+Toilet2	CKT2	SB2	TL10	40	0.74	C1,C2,C3,C4	10A
			TL2	40			
			LL2	15			
			F2	30			
			F3	30			
			S2		5	C9	
			Total		5.74		

Master Bedroom+ Toilet 3 + Verandah	CKT3	SB3	TL3	40	0.81	C1,C2,C3	10A
			TL4	40			
			LL3	15			
			LL4	15			
			F4	30			
			F5	30			
			S3		5	C9	
			Total		5.81		
Guest room	CKT4	SB4	TL5	40	0.66	C1,C2	10A
			TL6	40			
			F5	30			
			F6	30			
			S4		5	C9	
			Total		5.66		
Living space 1	CKT5	SB6	TL9	40	0.33	C1,C2,C3,C4	10A
			F10	30			
			S5		5	C9	
			Total		5.33		
Living space 2+ Common Toilet	CKT6	SB5	TL7	40	0.88	C1,C2	10A
			TL8	40			
			F7	30			
			F8	30			
			F9	30			
			Total		5.88		
Kitchen	CKT7	SB7	LL7	15	0.33	C1	10A
			LL8	15			
			F	40			
			S6		5	C9	
			Total		5.33		

Emergency Sub-Distributin Board (ESDB) -1 Fixtures							
Room Name	Circuit No	Switchboard	Fixture	Power (W)	Current Rating(A)	Wire Rating	Breaker to SDB
Bedroom 1+Toilet 1	CKT1	SB1	TL1	40	0.26	C1	5A
			LL1	15			
			Total		0.26		
Bedroom 2+Toilet2	CKT2	SB2	LL2	15	0.26	C1	5A
			TL2	40			
			Total		0.26		
Master Bedroom+ Toilet 3 + Verandah	CKT3	SB3	TL3	40	0.40	C1	5A
			LL3	15			
			F5	30			
			Total		0.40		
Guest room	CKT4	SB4	TL5	40	0.33	C1	5A
			F6	30			
			Total		0.33		
Living space 2+ Common Toilet	CKT6	SB5	TL8	40	0.40	C1	5A
			F8	30			
			Total		0.40		
Kitchen	CKT7	SB7	LL7	15	0.07	C1	5A
			Total		0.07		

Basement Sub-Distributin Board (B_SDB) -1 Fixtures							
Room Name	Circuit No	Switchboard	Fixture	Power (W)	Current Rating(A) $I=P/VCOS \theta$	Wire Rating	Breaker to SDB
Gym	Ckt1	SB1	TL(1-16)	(16*40)=640	3.06	C1	5A
				TOTAL			
Parking area	Ckt2	SB2	TL(17-19)	(3*40)=120	0.57	C1	10A
				S	5	C9	
				TOTAL	5.57		

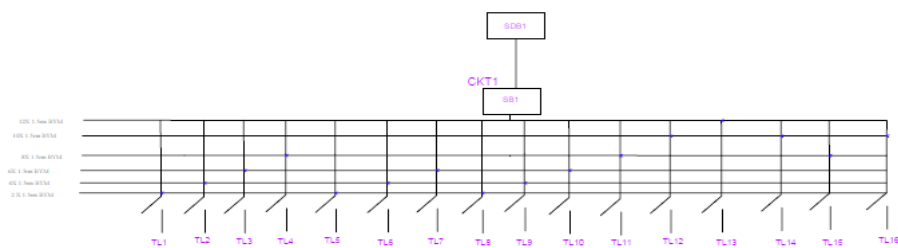
Basement Emergency Sub-Distributin Board (B_ESDB) -1 Fixtures							
Room Name	Circuit No	Switchboard	Fixture	Power (W)	Current Rating(A) $I=P/VCOS \theta$	Wire Rating	Breaker to SDB
Parking area	Ckt1	SB1	TL3,TL6,TL12,TL8	(4*40)=160	0.77	C1	10A
				S	5	C9	
				TOTAL	5.77		

GROUND FLOOR SDB Fixture

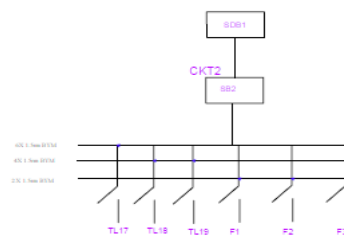
Room Name	Circuit No	Switchboard	Fixture	Power (W)	Current Rating(A) $I=P/VCOS \theta$	Breaker to SDB
Ground full	CKT1	SB1	TL(1-9)	(9*40)=360	2	5A
			F1	3		
			LL1	15		
			LL2	15		
			Total	420		

GROUND FLOOR SDB Emergency Fixture						
Room Name	Circuit No	Switchboard	Fixture	Power (W)	Current Rating(A) $I=P/VCOS \theta$	Breaker to SDB
Ground full	CKT1	SB1	TL(1)	40W	0.19	5A

Switch Board connection DiagramBasement

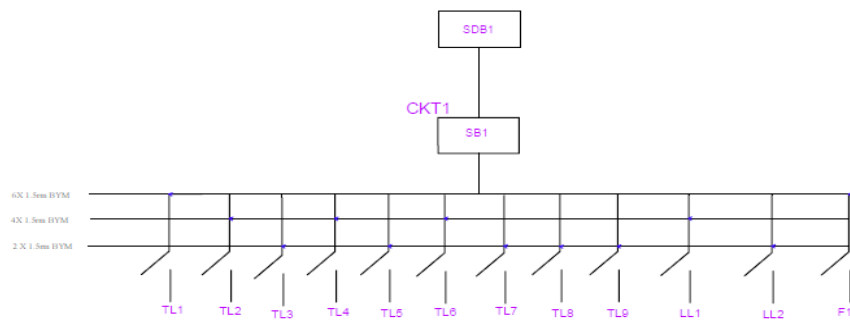


Gym

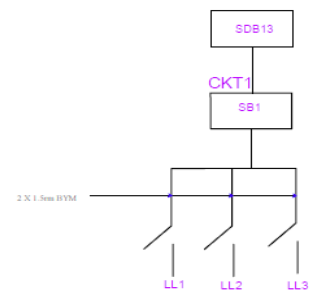


Switch Board connection Diagram

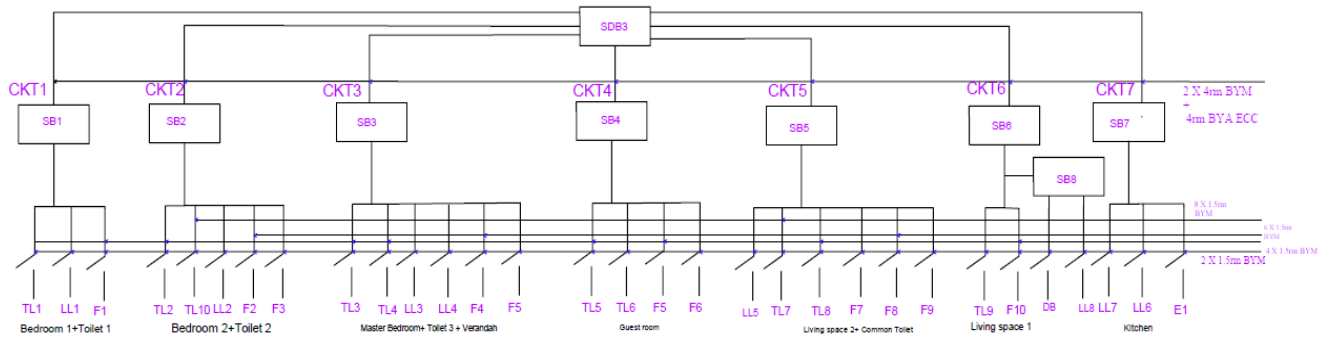
Ground



Rooftop

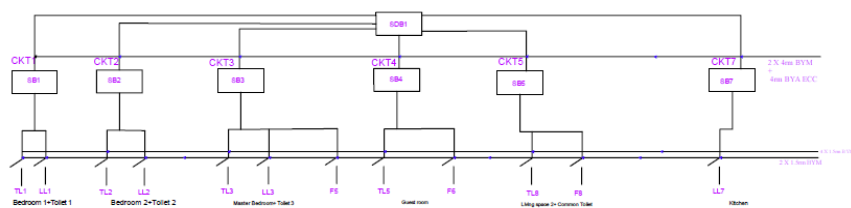


Switch Board Connection Diagram(Per Flat)



Emergency switch Board connection Diagram

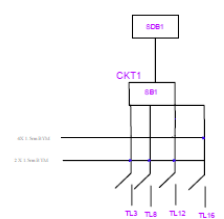
Emergency Switch Board Connection Diagram Per Flat



Ground



Basement



Sub-Distribution Board Calculation

GROUND:

Ckt breaker needed: 5 A. (total current 2A)

BASEMENT:

CKT1=3.06A

CKT2=5.59A

S1=20A

Total current=28.65A

Ckt breaker needed:30 A.

(TYPICAL FLOOR PER FLAT):

Total Power (Light Load): P_{LL}=7875W

P_{HL(20A)}= 18 KW

$$P_{\text{total}} = (P_{LL} * 0.7) + (P_{HL_20A} * 0.2) \\ = 9112.5\text{w}$$

$$P_{\text{total}} = v * I * \cos\theta$$

$$I = P_{\text{total}} / v * \cos\theta$$

$$= 9112.5 / 220 * 0.95$$

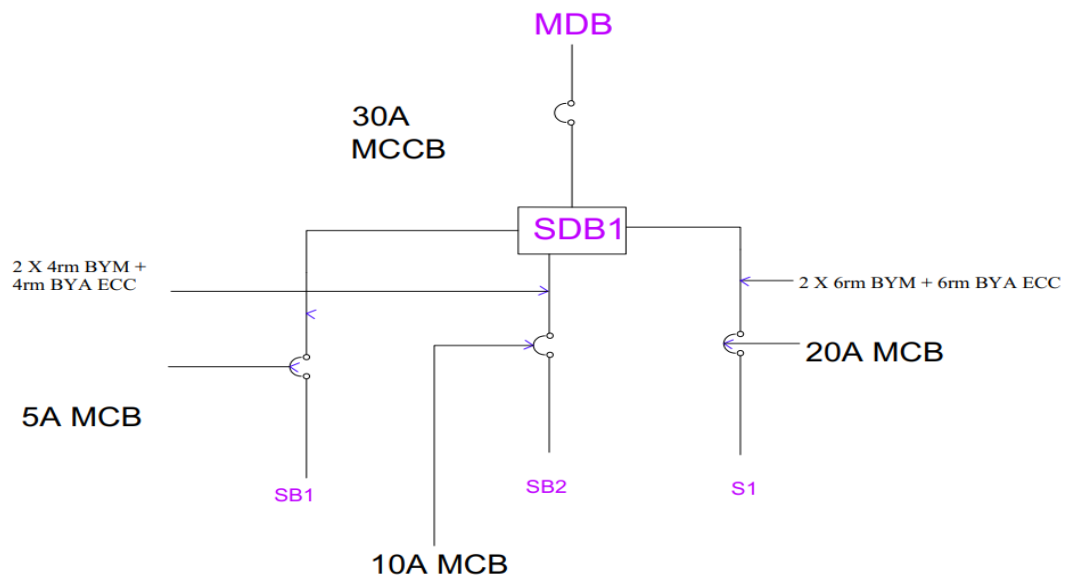
$$= 43.6\text{A}$$

Ckt breaker needed:45 A.

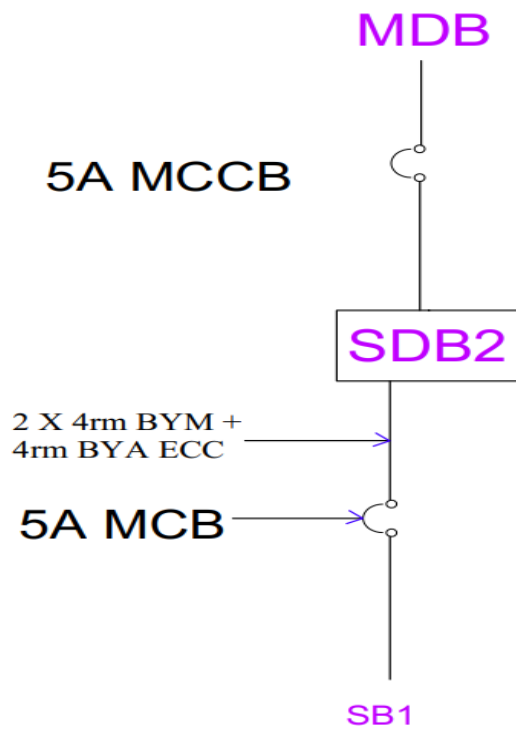
SDB SUMMARY:

	Light	Fan	5A socket	Total SB power			15A socket	20A Socket	P_total	
Bedroom_1 + Toilet 1	55	30	1000	1085			0	3000	18225	
Bedroom_2 + Toilet 2	95	60	1000	1155			0	3000		
Master Bedroom + Toilet 3+ Veranda	110	60	1000	1170			0	3000		
Guest Room	80	60	1000	1140			0	3000		
Dinning space and living space 1	40	30	1000	1070			0	3000		
g space and living space 2 + Common	95	90	1000	1185			0			
Kitchen	30	40	1000	1070			0	3000		
	Light load power (1 flat)>>>>>>			7875		15A per flat>>>	0	18000	<<< 20A per flat	
	Total PLL (2 flat)>>>>>>>>			15750		PHL 15A>>>	0	36000	<<<PHL 20A	
	Here, The Fan is assumed 30W, Exhaust Fan is 40W									
	5A socket is 1000W									
	20A socket 3000W									

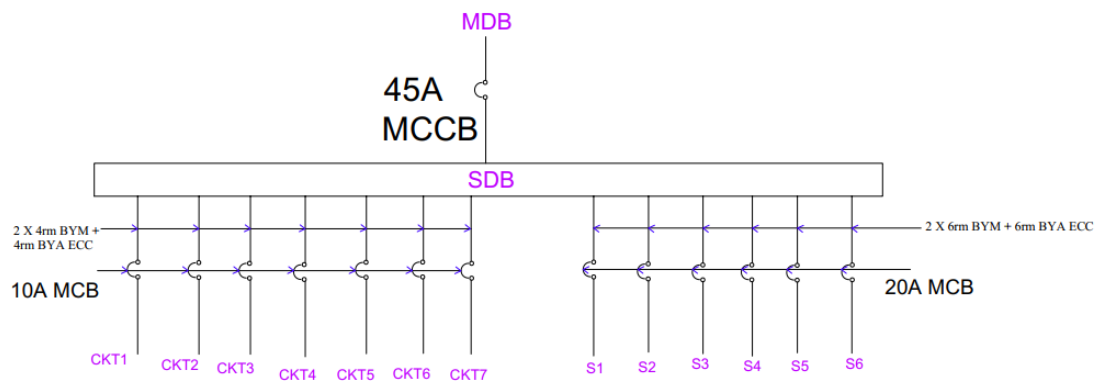
Basement SDB Diagram



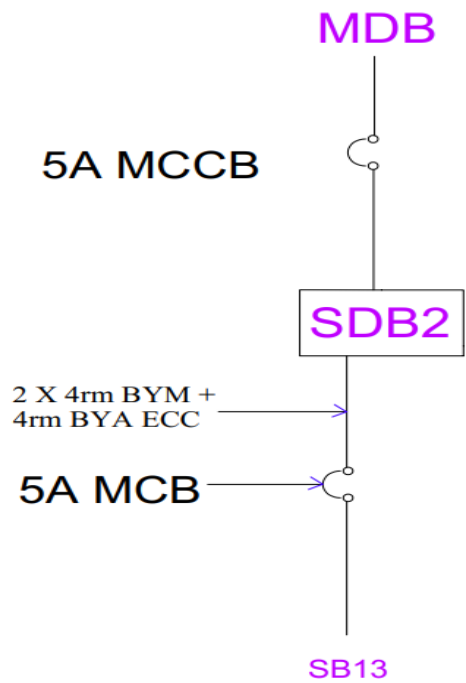
Ground SDB Diagram



Typical floor SDB Diagram

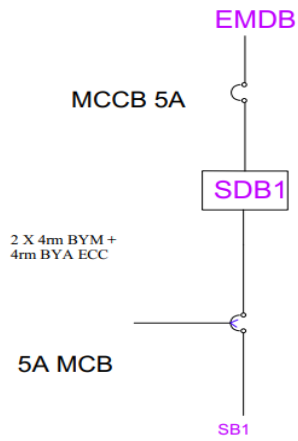


Roof Top SDB Diagram

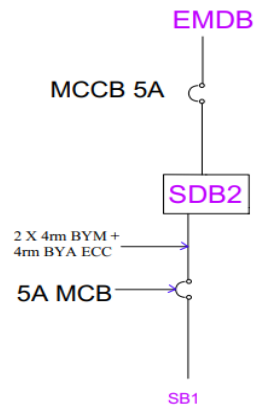


Sub-Distribution Board Emergency

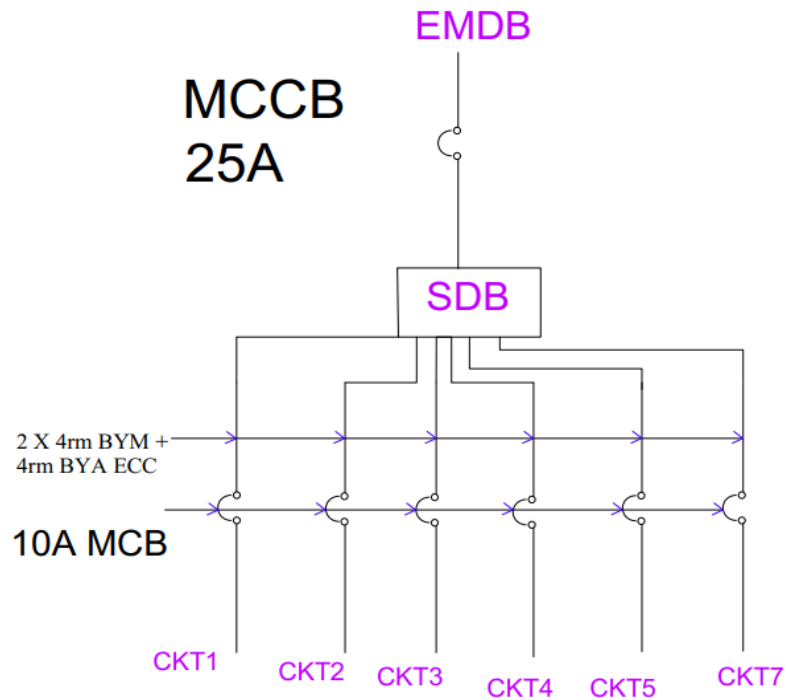
Basement SDB Diagram



Ground SDB Diagram



Typical floor SDB Diagram



Main-Distribution Board Calculation

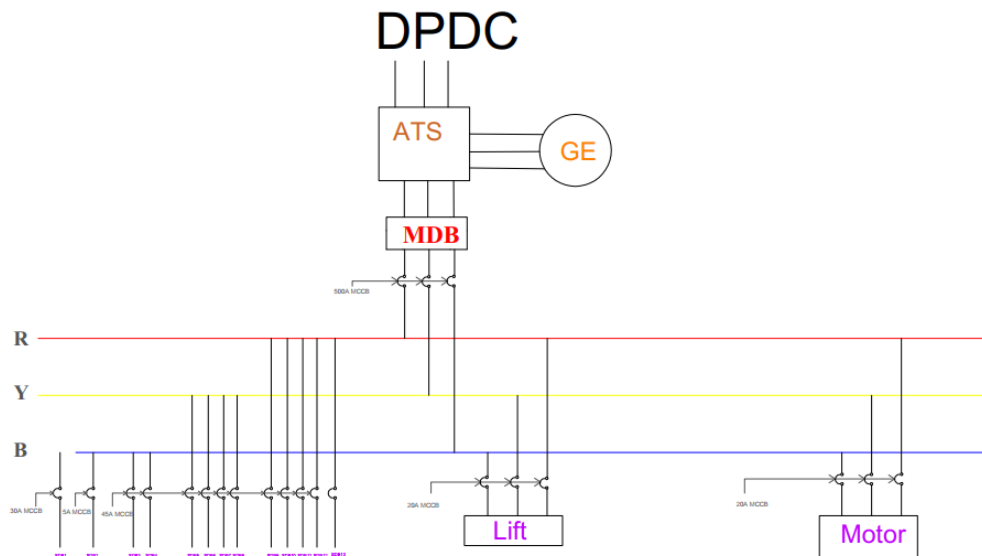
SDB1=30A

SBD2=5A

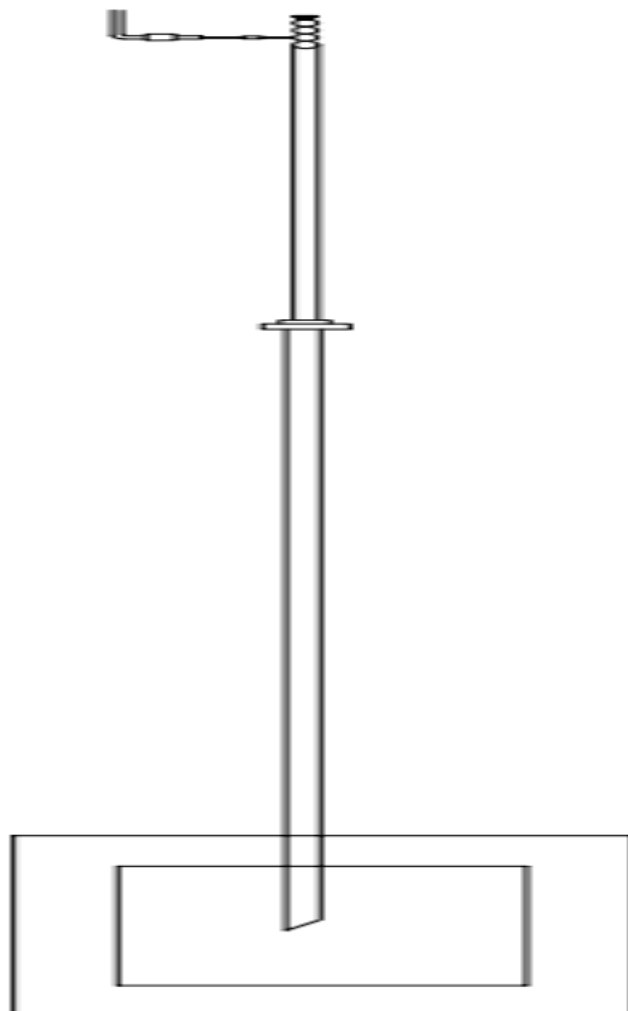
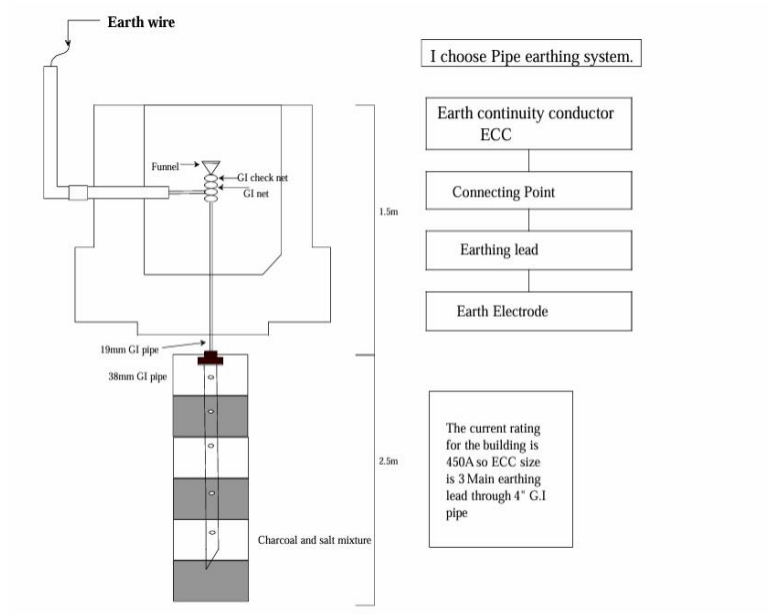
SBD(Typical)=45*10=450A

SDB13=5A

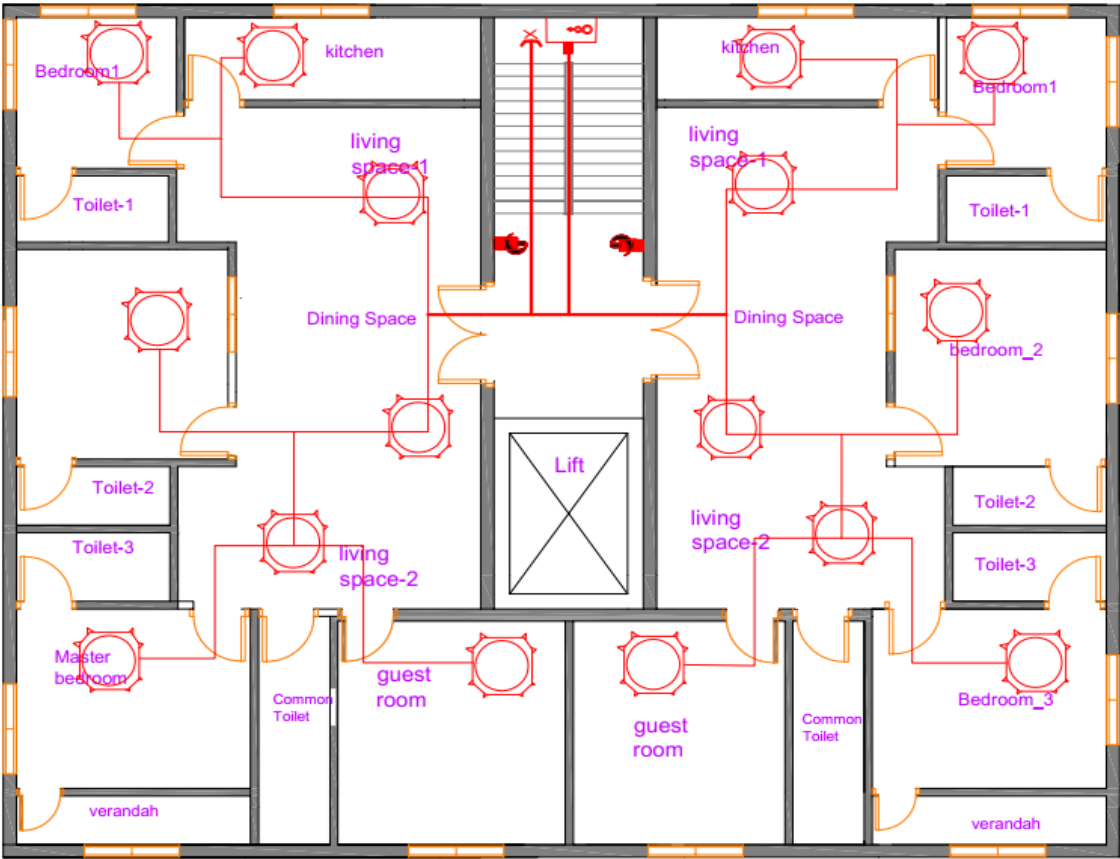
Total SDB=30+5+450+5=490A



Earthing System of the Building



Fire System



Conclusion

In this project, we have performed the designing of a three-storey building floor plan along with the electrical fixtures and conduit layout. Then, we have designed the switchboard connection diagram showing how the incoming electric power is distributed throughout the residential building. Along with the general connectivity, different wire schedules and protection equipment such as circuit breakers have been shown in the single line diagrams. To protect the building from electrical surges caused by lightning strike, we have planned the necessary lightning protection system. Thus, we have gained a hands-on experience on the electrical service design of a residential building.

Acknowledgments

We would like to thank Nure Alam Dipu Sir , Dr. for their valuable guidance in this project.

References Book List

References [1]

Table 8.1.5: Recommended Values of Illumination for Residential Buildings

Area or Activity	Illuminance (lux)	Area or Activity	Illuminance (lux)
Dwelling Houses		Hotels	
Bedrooms		Entrance halls	150
General	70	Reception and accounts	300
Bed-head, Dressing table	250	Dining rooms (tables)	150
Kitchens	200	Lounges	150
Dining rooms (tables)	150	Bedrooms	
Bathrooms		General	100
General	100	Dressing tables, bed heads, etc.	250
Shaving, make-up	300	Writing rooms (tables)	300
Stairs	100	Corridors	70
Lounges	100	Stairs	100
Garages & Porches	100	Laundries	200

Area or Activity	Illuminance (lux)	Area or Activity	Illuminance (lux)
Basement Car Park	100	Kitchens	
Porches, Entrances	70	Food stores	100
Sewing and darning	600	Working areas	250
Reading (casual)	150	Goods and passenger lifts	70
Home work and sustained reading	300	Cloak-rooms and toilets	100
		Bathrooms	100
		Above mirror in bathrooms	300

References [2]

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Course No. EEE-230

Table for Cables, Conduits, ECC, EL, Voltage drop and Current ratings of different specifications as per Manual of Eastern Cables, BICC cables and Tables, Electrical Conductors (International Standard Sizes) etc. :

A	B	C	D	E	F		G	H	I		J	
					a'	b'			a"	b"	a'''	B'''
3/0.029	1.5	5	16	10	6	10		27	27	22	16	20
7/0.029	2.5	10	16	10	4	7		16	36	30	22	28
7/0.036	4	15	14	10	3	5	1	10	47	39	30	37
7/0.044	6	20	14	10	2	4	1	6.8	59	50	38	47
7/0.052	10	30	10	10	1	2	1.5	4	78	68	52	63
7/0.064	16	40	10	10		1	1.5	2.6	100	94	70	85
19/0.052	25	50	6	6		1	2	1.6	130	125	91	110
19/0.064	35	60	6	6			2	1.2	155	160	112	136
19/0.072	50	70	6	6			2	0.93	185	195	136	164
19/0.083	70	100	1/0	1/0			2	0.65	225	245	173	207
37/0.072	95	120	1/0	1/0			2.5	0.48	270	300	216	253
37/0.083	120	150	1/0	1/0			2.5	0.4	310	350	244	291
37/0.093	150	200	1/0	1/0			3	0.34	350	405		333
37/0.130	185	250	3/0	3/0			3.5	0.29	390	460		381
61/0.093	240	300	3/0	3/0			4	0.24	450	555		452
61/0.103	300	425	3/0	3/0			4	0.22	515	640		526
91/0.093	400	585	3/0	3/0			6	0.2	586	770		639
91/0.103	500	685	3/0	3/0			6	0.18	680	900		752
127/0.103	630	800	3/0	3/0			6	0.17	800	1030		855

A : Single core cable construction diameter, inch as per Imperial Standard Size : B.S.S (old).

B : Single core cable construction area, mm² as per Metric Standard Size : VDE.

C : CB designed current rating amps.

D : ECC (Earth Continuity Conductor), SWG.

E : EL (Earthing Lead), SWG

F : No. of cables in

a') 3/4" diameter conduit

b') 1" diameter conduit

G : GI pipe diameter (for 4 – core cable), inch.

H : Volt drop /amp/meter, Vd in mV (For PVC insulated, non-armoured single core cable 600/1000 volts as per BICC Metric Supplement, page 20-22, September 1969).

I : Maximum Current rating (For Type : NYY to VDE 0271/3, 69)

a") 30° C ambient temperature, underground, amps

b") 35° C ambient temperature in air, amps

J : Maximum current carrying capacity (For Type : BYA to B.S. 6004 : 1975)

a") Bunched & Enclosed in conduit, two cables single phase at 35° C, amps

b") Clipped to a surface or on a cable tray bunched and un-enclosed two cables single phase at 35° C, amps

NYY : PVC insulated and PVC sheathed cable, rated voltage 600/1000 volts.

BYA : PVC insulated non-sheathed single core cable, rated voltage 450/750 volts.

References [3]

1.3.33 Lightning Protection of Buildings

Whether a building needs protection against lightning depends on the probability of a stroke and acceptable risk levels. Assessment of the risk and of the magnitude of the consequences needs to be made. As an aid to making a judgment, a set of indices is given in Table 8.1.27 below for the various factors involved.

Table 8.1.27 (a): Index Figures Associated with Lightning Protection Design

Index A: Use of Structure	Index	Index B: Type of Construction	Index
Houses and similar buildings	2	Steel framed encased with nonmetal roof ^a	1
Houses and similar buildings with outside aerial	4	Reinforced concrete with nonmetal roof	2
Small and medium size factories, workshops and laboratories	6	Brick, plain concrete, or masonry with nonmetal roof	4
Big industrial plants, telephone exchanges, office blocks, hotels, blocks of flats	7	Steel framed encased or reinforced concrete with metal roof	5
Places of assembly, for example, places of workshop, halls, theatres, museums, exhibitions, department stores, post offices, stations, airports, stadiums	8	Timber formed or clad with any roof other than metal or thatch	7
Schools, hospitals, children's homes and other such structures	10	Any building with a thatched roof	10

^a A structure of exposed metal which is continuous down to ground level is excluded from the table as it requires no lightning protection beyond adequate earthing arrangements.

Table 8.1.27 (b): Index Figures Associated with Lightning Protection Design

Index C : Contents or Consequential Effects	Index	Index D : Degree of Isolation	Index
Ordinary domestic or office building, factories and workshops not containing valuable materials	2	Structure located in a large area having structures or trees of similar or greater height, e.g. a large town or forest	2
Industrial and agricultural buildings with specially susceptible ^b contents	5	Structure located in an area with a few other structures or trees of similar height	5
Power stations, gas works, telephone exchanges, radio stations	6	Structure completely isolated or exceeding at least twice the height of surrounding structures or trees	10
Industrial key plants, ancient monuments, historic buildings, museums, art galleries	8	Index E : Type of Terrain	Index
Schools, hospitals, children's and other homes, places of assembly	10	Flat terrain at any level	2
^b This means specially valuable plant or materials vulnerable to fire or the results of fire.		Hilly terrain	6
		Mountainous terrain 300 m and above	8

Table 8.1.27 (c) : Index Figures Associated with Lightning Protection Design

Index F : Height of Structure	Index	Index G : Lightning Prevalence	Index
Up to 9 m	2	Number of thunderstorm days per year:	
9-15 m	4	Up to 3	2
15-18 m	5	4-6	5
18-24 m	8	7-9	8
24-30 m	11	10-12	11
30-38 m	16	13-15	14
38-46 m	22	16-18	17
46-53 m ^c	30	19-21	20
^c Structures higher than 53 m require protection in all cases		Over 21	21